

From Kaprekar Dynamics to Logit Geometry: Exact Compression, Weighted Basins, and Limits of a Sorted-Logit Representation

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Abstract

The Kaprekar routine is a finite digit-rearrangement dynamical system whose best-known decimal fixed point is 6174. Outer-digit difference coordinates for this routine are classical. Starting from that prior art, this paper gives an explicit integer linear reconstruction, proves that it is injective, and derives the exact one-step image size

$$|\text{Im } K_{b,n}| = \binom{b + \lfloor n/2 \rfloor - 1}{b - 1}.$$

The restriction of the full map to its one-step image is conjugate to a reduced map on spectra. With exact spectrum-fiber weights, this conjugacy recovers the complete functional graph, raw-state basin sizes, hitting-time histograms, and entropy evolution in polynomial time for every fixed base. Certified decimal computations show that nontrivial cycles, rather than fixed points, capture most raw inputs at widths five through eight.

The same adjacent-gap operator yields normalized coordinates for sorted top- k language-model logits. We prove reconstruction, singularity at a flat spectrum, near-flat sensitivity, change-of- k behavior, sharp omitted-tail bounds, and a distribution-free calibration impossibility. These coordinates are exactly an invertible reparameterization of relative sorted logits; no predictive improvement follows from the reparameterization alone. We also derive the exact accept-and-residual condition for distribution-preserving speculative sampling and show why a target-only rank or margin gate is not exact in general. An independent integer-arithmetic checker exhaustively verifies 49 finite systems comprising 3,816,497 raw states, while a separate Lean development mechanizes the core pairing identities and part of the counting argument. No empirical large-language-model performance claim is made.

Keywords: Kaprekar routine; finite dynamical systems; functional graphs; exact basin weights; entropy; sorted logits; uncertainty estimation; speculative decoding.

1 Introduction

Kaprekar’s digit-rearrangement routine begins with a fixed-width digit string, sorts its digits in descending and ascending order, and subtracts the latter number from the former. The decimal fixed points 495 and 6174 made the routine widely known [11, 12]. The broader mathematical object is not a single constant, however, but a family of finite functional graphs indexed by the base and the width. Longer widths and other bases exhibit multiple fixed points, genuine cycles, and nonuniform basins [9, 20, 23, 13, 14].

The first purpose of this paper is to make a precise compression theorem for these graphs. Prichett, Ludington, and Lapenta already observed that a Kaprekar transform depends entirely

on ordered outer-digit differences [20]; this dependence is therefore not claimed as new. The contribution developed here is the chain of consequences obtained from an explicit linear map and its injectivity: an exact image count, a conjugacy after the first step, reconstruction of the full functional graph from a weighted reduced graph, polynomial fixed-base basin computation, and an exact entropy pushforward identity.

The second purpose is to delimit a proposed analogy with language-model logits. Sorted top- k logits can be represented by normalized adjacent gaps. This is a clean coordinate system, but it is an invertible reparameterization of relative sorted logits, not a source of additional information. It also discards token identity and, unless supplied separately, tail information. Accordingly, this paper proves only structural identities and impossibility results for these coordinates. It does not report or imply a gain in hallucination detection, selective prediction, routing, or decoding speed.

The main contributions are:

1. an explicit factorization $K_{b,n} = \Lambda_{b,n} \circ \Sigma_{b,n}$, a superincreasing proof of injectivity, and an exact image-cardinality theorem valid also for width one;
2. an exact conjugacy on the one-step image and a reconstruction theorem for the entire raw functional graph using spectrum-fiber weights;
3. exact raw basin, depth, and entropy formulas, together with certified decimal results showing cycle-dominated dynamics at widths five through eight;
4. a corrected account of normalized sorted-logit gaps, including their exact information content, degeneracies, sensitivity, tail limits, and distribution-free limitations; and
5. an exact acceptance/residual theorem for speculative sampling and a reproducibility package with independent exhaustive certificates and partial formal verification.

Evidence discipline. Universal algebraic results, finite computational certificates, and empirical hypotheses are kept separate. Exhaustive computation can prove a declared finite instance and can falsify an implementation, but it is not substituted for proofs over infinitely many (b, n) . Conversely, theorems about a coordinate system do not establish predictive value on real model outputs.

2 Prior work and novelty boundary

2.1 Kaprekar dynamics

The classical literature classified four-digit constants and decimal fixed points [9, 20]. The 1981 paper explicitly introduced arbitrary-base outer-difference tuples and stated that the transformation depends entirely on them. Later work studied higher bases, symmetries, fixed points, cycles, and convergence distances [23, 6, 18, 13, 14]. Iwasaki gave an explicit modern classification of decimal nonzero fixed points and clarified the distinction between a fixed point and a globally attracting Kaprekar constant [10]. Dahl analyzed attractor-basin entropy and coarse-grained drift fields for decimal systems [5].

Very recent work by Chen, Ono, Schwartz, and Thakur gives a much stronger number-theoretic classification for four-digit odd bases: after a bounded transient, the stable dynamics are conjugate to projective doubling [3]. That specialized theorem and its separate Lean artifact should not be replaced by generic enumeration. The present results are complementary: they hold for arbitrary base and width and expose exact image, graph, and weight structure, but do not reproduce the odd-base classification.

The defensible novelty claim is therefore deliberately narrow. We do not claim the discovery of outer-gap reduction. We claim the explicit superincreasing encoding, exact image cardinality, post-first-step conjugacy, complete weighted-graph reconstruction, and their information-theoretic consequences.

2.2 Logit uncertainty and speculative decoding

Output-side uncertainty estimates coexist with semantic and internal-state methods [1, 8, 15, 17]. Recent evidence shows that entropy-only selective policies can fail in model-dependent ways and that hybrid scores can improve measured risk-coverage behavior on some benchmarks [19]. Other direct baselines include logit-magnitude methods [25] and local sorted-logit geometry such as Min- k sampling [7]. These works make an information-matched raw relative-logit baseline mandatory for any future experiment on the coordinates studied here.

Exact speculative decoding uses modified rejection sampling to preserve the target distribution [16, 2]. Margin-aware relaxed verification has already been studied at model scale by MARS [22], and recent acceptance-certificate theory gives sharper criteria for several deterministic and relaxed rules [21]. Consequently, the margin identity in this paper is presented as an elementary diagnostic, not as a novel distribution-preserving decoder.

3 Kaprekar maps and ordered outer gaps

Fix a base $b \geq 2$ and a width $n \geq 1$. Let

$$X_{b,n} = \{0, 1, \dots, b^n - 1\},$$

where every element is represented by exactly n base- b digits, including leading zeros. If the digits sorted in descending order are

$$a_1 \geq a_2 \geq \dots \geq a_n,$$

define the fixed-width Kaprekar map

$$K_{b,n}(x) = \sum_{i=1}^n a_i b^{n-i} - \sum_{i=1}^n a_i b^{i-1}. \quad (1)$$

Set $m = \lfloor n/2 \rfloor$.

Definition 3.1 (Ordered outer-gap spectrum). For $1 \leq j \leq m$, define

$$\Delta_j = a_j - a_{n+1-j}.$$

The spectrum map is $\Sigma_{b,n}(x) = (\Delta_1, \dots, \Delta_m)$, with codomain

$$S_{b,n} = \{\Delta \in \mathbb{Z}^m : b-1 \geq \Delta_1 \geq \dots \geq \Delta_m \geq 0\}.$$

For $n = 1$, $S_{b,1}$ contains the unique empty spectrum.

Every actual spectrum lies in $S_{b,n}$. The coordinate bounds are immediate, and for $j < m$,

$$\Delta_j - \Delta_{j+1} = (a_j - a_{j+1}) + (a_{n-j} - a_{n+1-j}) \geq 0.$$

Definition 3.2 (Linear reconstruction). For $1 \leq j \leq m$, let

$$c_j = b^{n-j} - b^{j-1}, \quad \Lambda_{b,n}(\Delta) = \sum_{j=1}^m c_j \Delta_j.$$

For $m = 0$, the empty sum is zero.

Theorem 3.3 (Exact factorization). For every $x \in X_{b,n}$,

$$K_{b,n}(x) = \Lambda_{b,n}(\Sigma_{b,n}(x)).$$

Proof. In (1), pair the term indexed by j with the term indexed by $n+1-j$. Their combined contribution is

$$(a_j - a_{n+1-j})(b^{n-j} - b^{j-1}) = c_j \Delta_j.$$

If n is odd, the unpaired middle coefficient is zero. Summing the m pairs proves the identity. $\square$

Lemma 3.4 (Spectrum surjectivity). The map $\Sigma_{b,n} : X_{b,n} \rightarrow S_{b,n}$ is surjective.

Proof. For any $\Delta \in S_{b,n}$, use the sorted length- n digit string

$$(\Delta_1, \dots, \Delta_m, 0, \dots, 0).$$

Its outer differences are exactly $\Delta_1, \dots, \Delta_m$. When $n = 1$, every raw digit has the unique empty spectrum. $\square$

Lemma 3.5 (Superincreasing coefficients). For every $1 \leq j \leq m$,

$$c_j > (b-1) \sum_{k=j+1}^m c_k.$$

Proof. Finite geometric summation gives

$$c_j - (b-1) \sum_{k=j+1}^m c_k = b^{n-m} + b^m - b^j - b^{j-1}.$$

Because $j \leq m$ and $n-m \geq m$,

$$b^j + b^{j-1} \leq b^m + b^{m-1} < 2b^m \leq b^{n-m} + b^m.$$

The difference is therefore strictly positive. $\square$

Theorem 3.6 (Injectivity and exact image size). The map $\Lambda_{b,n}$ is injective on $S_{b,n}$, and

$$|\text{Im } K_{b,n}| = |S_{b,n}| = \binom{m+b-1}{b-1} = \binom{b + \lfloor n/2 \rfloor - 1}{\lfloor n/2 \rfloor}. \quad (2)$$

Proof. For distinct Δ, Δ' , let j be their first differing coordinate and orient them so that $\Delta_j - \Delta'_j \geq 1$. Every later coordinate difference has magnitude at most $b-1$, so Lemma 3.5 implies

$$\Lambda(\Delta) - \Lambda(\Delta') \geq c_j - (b-1) \sum_{k>j} c_k > 0.$$

Thus Λ is injective. Theorem 3.3 and Lemma 3.4 give $\text{Im } K = \Lambda(S_{b,n})$.

To count spectra, let q_r be the number of coordinates of Δ equal to r , for $r = 0, \dots, b-1$. The weakly decreasing spectrum is uniquely determined by the weak composition

$$q_0 + q_1 + \dots + q_{b-1} = m.$$

Stars and bars gives $\binom{m+b-1}{b-1}$. This argument also covers $m = 0$, where there is one empty spectrum. $\square$

Corollary 3.7 (Fixed-base compression). *For fixed b ,*

$$|\text{Im } K_{b,n}| = \frac{m^{b-1}}{(b-1)!} + O(m^{b-2}) = \Theta(n^{b-1}),$$

whereas $|X_{b,n}| = b^n$. Hence the one-step image ratio tends to zero exponentially in n .

Remark 3.8. The spectrum has tuple length m , but at fixed base it can be stored using the b histogram counts $q_0, \dots, q_{b-1}$ with one sum constraint. It is therefore misleading to call m an unqualified effective dimension.

4 Exact conjugacy and complete functional graphs

Write $\Sigma = \Sigma_{b,n}$, $\Lambda = \Lambda_{b,n}$, $K = K_{b,n}$, and define

$$R = \Sigma \circ \Lambda : S_{b,n} \rightarrow S_{b,n}, \quad I = \text{Im } K = \Lambda(S_{b,n}).$$

Theorem 4.1 (Post-first-step conjugacy). *The map $\Lambda : S_{b,n} \rightarrow I$ is a bijection satisfying*

$$K|_I \circ \Lambda = \Lambda \circ R.$$

Consequently $K|_I$ and R have corresponding cycles with identical least periods, and for every $x \in X_{b,n}$ and $t \geq 1$,

$$K^t(x) = \Lambda R^{t-1} \Sigma(x). \quad (3)$$

Proof. Theorem 3.6 makes Λ a bijection onto I . Direct composition gives

$$K\Lambda = \Lambda\Sigma\Lambda = \Lambda R.$$

This is the conjugacy identity, and (3) follows by induction. Bijective conjugacy preserves least periods. $\square$

Semiconjugacy alone would not justify exact cycle correspondence; injectivity is the missing step. Every raw orbit enters I after one application of K .

For a spectrum Δ , define its raw fiber weight

$$w_{b,n}(\Delta) = |\Sigma^{-1}(\Delta)|.$$

Theorem 4.2 (Complete graph from a weighted reduced graph). *For every $\Delta \in S_{b,n}$,*

$$\text{indeg}_K(\Lambda\Delta) = w_{b,n}(\Delta).$$

The entire functional graph of K is recovered, up to the integer labels of anonymous leaves, by relabeling each reduced vertex Δ as $\Lambda\Delta$ and attaching exactly

$$w_{b,n}(\Delta) - \text{indeg}_R(\Delta)$$

additional indegree-zero source leaves to it. Every state outside I has indegree zero.

Proof. Injectivity gives

$$K(x) = \Lambda\Delta \iff \Lambda\Sigma(x) = \Lambda\Delta \iff \Sigma(x) = \Delta.$$

Thus the full indegree equals the fiber size. A preimage already in I has the form $\Lambda\Delta'$ and maps to $\Lambda\Delta$ exactly when $R(\Delta') = \Delta$. These are the $\text{indeg}_R(\Delta)$ reduced predecessors. All other preimages lie outside I ; by definition of $I = \text{Im } K$, such a state cannot itself have a predecessor. $\square$

The theorem reconstructs graph structure and multiplicities. Recovering the original integer label of every source leaf still requires enumerating or otherwise describing its spectrum fiber.

5 Exact fiber weights, basins, and information loss

Let $a = (a_1, \dots, a_n)$ be a sorted digit multiset and let $c_d(a)$ count the occurrences of digit d . The number of fixed-width strings represented by a is

$$\text{mult}(a) = \frac{n!}{\prod_{d=0}^{b-1} c_d(a)!}.$$

Therefore

$$w_{b,n}(\Delta) = \sum_{\substack{b-1 \geq a_1 \geq \dots \geq a_n \geq 0 \\ a_j - a_{n+1-j} = \Delta_j}} \frac{n!}{\prod_{d=0}^{b-1} c_d(a)!}. \quad (4)$$

Theorem 5.1 (Exact raw basin weight). *Let C be a cycle of R , and let*

$$\text{Basin}_R(C) = \{\Delta : R^t(\Delta) \in C \text{ for some } t \geq 0\}.$$

Then $\Lambda(C)$ is the corresponding full cycle and

$$|\text{Basin}_K(\Lambda C)| = \sum_{\Delta \in \text{Basin}_R(C)} w_{b,n}(\Delta).$$

Proof. The spectrum fibers partition $X_{b,n}$. By Theorem 4.1, a raw state reaches $\Lambda(C)$ exactly when its spectrum reaches C . Summing precisely those fiber sizes proves the formula. $\square$

Corollary 5.2 (Exact hitting-time histogram). *If $\Delta \notin C$ has reduced distance $d_R(\Delta, C)$, every state in its fiber has raw hitting time $d_R(\Delta, C) + 1$. If $\Delta \in C$, exactly one state in its fiber is already on $\Lambda(C)$ and the remaining $w_{b,n}(\Delta) - 1$ states enter the cycle in one step.*

Proof. For a noncycle spectrum, (3) and injectivity show that first cycle entry occurs one step after its reduced entry. For $\Delta \in C$, the cyclic permutation $R|_C$ has a unique predecessor $\gamma \in C$. The state $\Lambda\gamma$ is the unique cycle member whose spectrum is Δ ; every other state in that fiber maps to $\Lambda\Delta$ in one step. $\square$

There are $\binom{b+n-1}{b-1}$ sorted digit multisets. For fixed b , this number and the reduced-state count are both $\Theta(n^{b-1})$. Each transition, multinomial weight, and graph traversal uses arithmetic on integers with bit length polynomial in n . Equations (4) and Theorem 5.1 therefore give a polynomial-time exact algorithm in the standard bit model for every fixed base, without enumerating all b^n strings.

Theorem 5.3 (Entropy pushforward). *Let X have any distribution μ on $X_{b,n}$ and let $\nu = \Sigma_*\mu$. For every $t \geq 1$,*

$$(K^t)_*\mu = \Lambda_*(R^{t-1})_*\nu, \quad H(K^t(X)) = H(R^{t-1}(\Sigma X)).$$

Under a uniform raw input,

$$\Pr(\Sigma X = \Delta) = \frac{w_{b,n}(\Delta)}{b^n}$$

and the exact one-step Shannon information loss is

$$H(X \mid K(X)) = \sum_{\Delta \in S_{b,n}} \frac{w_{b,n}(\Delta)}{b^n} \log_2 w_{b,n}(\Delta). \quad (5)$$

It obeys

$$H(X \mid K(X)) \geq n \log_2 b - \log_2 \binom{m+b-1}{b-1}.$$

Proof. The pushforward identity is the random-variable form of (3). The injective relabeling Λ preserves atom probabilities and hence discrete entropy. Under the uniform law, conditioning on $K(X) = \Lambda\Delta$ leaves a uniform choice among exactly $w_{b,n}(\Delta)$ raw states, proving (5). Finally, K is deterministic, so $H(X \mid K(X)) = n \log_2 b - H(K(X))$, and entropy on a support of size $|\text{Im } K|$ is at most the logarithm of that size. $\square$

For fixed b , the support-based loss grows as

$$n \log_2 b - (b-1) \log_2 n + O(1).$$

This is a concrete negative engineering consequence. The map has a polynomial-size image and efficiently constructible collision fibers; it must not be represented as a cryptographic hash, entropy source, or reversible encoding.

6 Certified decimal dynamics

Table 1 reports exact raw basin sizes for decimal widths five through eight. The domain is all 10^n fixed-width strings, including leading zeros. Each width also has a ten-state repdigit basin leading to zero; that basin is omitted from the displayed nonzero rows.

Thus, at width five the two four-cycles absorb 96.8% of all raw inputs; at width six a seven-cycle absorbs 93.552%; at width seven the nonzero eight-cycle absorbs 99.9999%; and at width eight the three- and seven-cycles together absorb about 97.03%. A fixed-point-only summary therefore misses the dominant observed behavior at these widths. This does not contradict the special global role of 6174: among the decimal fixed-width strings of width four, the ten repdigits enter zero and the other 9,990 states reach 6174 in at most seven steps.

The table is a finite certified computation, not a universal theorem. The ordered transition and weight data are stored in `results/decimal_dynamics.json`; an independent oracle recomputes the cycles, weights, basin sizes, and hitting-time histograms before comparison.

Table 1: Exact nonzero decimal attractor basins under the uniform raw law.

Width	Attractor	Raw basin size	Share
5	2-cycle through 53955	3,190	3.190%
5	4-cycle through 63954	48,480	48.480%
5	4-cycle through 74943	48,320	48.320%
6	fixed point 549945	1,950	0.195%
6	fixed point 631764	62,520	6.252%
6	7-cycle through 851742	935,520	93.552%
7	8-cycle through 8429652	9,999,990	99.9999%
8	fixed point 63317664	599,536	0.599536%
8	fixed point 97508421	2,371,040	2.371040%
8	3-cycle through 86526432	48,247,316	48.247316%
8	7-cycle through 86326632	48,782,098	48.782098%

7 Normalized adjacent gaps for sorted logits

Let a vocabulary logit vector have size V , select its top $k \geq 2$ order statistics, and write

$$s_1 \geq s_2 \geq \cdots \geq s_k.$$

Assume first that $D = s_1 - s_k > 0$ and define normalized adjacent gaps

$$u_j = \frac{s_j - s_{j+1}}{D}, \quad j = 1, \dots, k-1. \quad (6)$$

These coordinates were called the Kaprekar gap simplex in the original proposal. Mathematically, they are normalized spacings of sorted logits.

Theorem 7.1 (Exact information content). *The vector u lies in*

$$\left\{ u \in \mathbb{R}_{\geq 0}^{k-1} : \sum_{j=1}^{k-1} u_j = 1 \right\},$$

a simplex of affine dimension $k-2$, and

$$s_i - s_k = D \sum_{j=i}^{k-1} u_j. \quad (7)$$

Consequently, u is a bijective coordinate for sorted top- k spectra modulo positive scaling and translation; (D, u) is a bijective coordinate modulo translation and exactly determines the top- k -conditional softmax law.

Proof. Nonnegativity follows from sorting. Telescoping gives

$$\sum_{j=1}^{k-1} u_j = \frac{s_1 - s_k}{D} = 1, \quad D \sum_{j=i}^{k-1} u_j = s_i - s_k.$$

The reconstruction formula proves the stated bijections. Softmax restricted and renormalized to the top k depends only on relative logits. $\square$

The exact connection with the digit construction is the shared interval-sum operator. For the applicable ranges of j ,

$$a_j - a_{n+1-j} = \sum_{i=j}^{n-j} (a_i - a_{i+1}), \quad s_j - s_{k+1-j} = D \sum_{i=j}^{k-j} u_i.$$

No stronger transfer theorem between the two application domains is claimed.

Define the translation-quotiented vector

$$r(s) = (s_1 - s_k, \dots, s_{k-1} - s_k, 0).$$

Corollary 7.2 (No information creation). *The observations $r(s)$ and (D, u) are bijectively related and therefore have identical Bayes-optimal risk for every target and loss. Since (D, u) is a deterministic function of the unrestricted sorted vector s , the latter can have lower Bayes risk. Equality for every decision problem is guaranteed, for example, by $Y \perp s_k \mid r(s)$.*

A varying common offset can predict Y while leaving (D, u) unchanged. An information-matched benchmark must therefore compare a same-capacity model on relative sorted logits. Any gain over that comparator can arise only from finite-sample inductive bias, regularization, numerical conditioning, or a capacity constraint, not from additional information.

7.1 Degeneracy, sensitivity, and the top-k choice

Proposition 7.3 (Flat-spectrum singularity). *For $k \geq 3$, the map in (6) has no continuous extension to $D = 0$.*

Proof. Choose distinct simplex points $u \neq v$. Reconstruct two sorted spectra with bottom logit zero and spread ε . Both approach the same all-zero vector as $\varepsilon \downarrow 0$, while their normalized coordinates remain u and v . A continuous extension would assign two limits to the same point. $\square$

For $k = 2$, $u = (1)$ whenever $D > 0$, so it contains no shape information. Implementations should expose a flat-spectrum flag rather than silently divide by an epsilon and report an ordinary simplex point.

Proposition 7.4 (Near-flat sensitivity). *Let sorted spectra s, t have spreads $D_s, D_t \geq d > 0$ and satisfy $\|s - t\|_\infty \leq \varepsilon$. Then*

$$\|u(s) - u(t)\|_\infty \leq \frac{4\varepsilon}{d}.$$

Proof. Each adjacent gap changes by at most 2ε , and the spread changes by at most 2ε . For corresponding gaps $0 \leq g' \leq D_t$,

$$\left| \frac{g}{D_s} - \frac{g'}{D_t} \right| \leq \frac{|g - g'|}{D_s} + \frac{g'|D_t - D_s|}{D_s D_t} \leq \frac{4\varepsilon}{d}.$$

$\square$

The $1/d$ factor captures unavoidable ill-conditioning near the flat stratum. Top- k token membership has an additional discontinuity at cutoff ties.

Proposition 7.5 (Exact change of k). *If a nondegenerate top- k spectrum is extended by $s_{k+1} \leq s_k$ and $D' = s_1 - s_{k+1}$, then*

$$u'_j = \frac{D}{D'} u_j \quad (1 \leq j < k), \quad u'_k = \frac{s_k - s_{k+1}}{D'}.$$

Thus changing k rescales every existing coordinate. Any experiment must predeclare k , sweep it, or use an explicitly multiscale representation.

7.2 Sharp omitted-tail bounds

Set

$$r_i = s_i - s_k = D \sum_{j=i}^{k-1} u_j, \quad A = \sum_{i=1}^k e^{r_i}.$$

For omitted logits $z_\ell \leq s_k$, define their partition mass relative to e^{s_k} by

$$T = \sum_{\ell > k} e^{z_\ell - s_k}.$$

Theorem 7.6 (Tail bounds). *For vocabulary size $V \geq k$,*

$$0 \leq T \leq V - k, \quad M_k = \frac{A}{A + T}, \quad p_1 = \frac{e^D}{A + T},$$

where M_k is the full-softmax top- k mass and p_1 is the full-softmax top-token probability. Therefore

$$\frac{A}{A + V - k} \leq M_k \leq 1, \quad \frac{e^D}{A + V - k} \leq p_1 \leq \frac{e^D}{A}.$$

The lower bounds are attained when every omitted logit equals s_k . If $V = k$, then $T = 0$ and (D, u) determines the complete softmax distribution. If $V > k$ and all logits are finite, the upper bounds are strict but sharp suprema as omitted logits tend to minus infinity.

Proof. Each of the $V - k$ tail summands is at most one and nonnegative. The formulas follow by dividing the full partition by e^{s_k} . Both M_k and p_1 are decreasing functions of T , proving the bounds and equality conditions. $\square$

When $V > k$, identical (D, u) values can have different tail mass, full entropy, and unconditional top-token probability. For example, give one tail token relative weight $t \in (0, 1]$. If P is the top- k -conditional law and $q = t/(A + t)$, the resulting entropy approaches

$$h_2(q) + (1 - q)H(P),$$

which is nonconstant in q . Recording the additional statistic

$$\tau = \text{logsumexp}(z_{\text{tail}}) - s_k$$

determines $T = e^\tau$ and all top- k full-softmax probabilities. For an empty tail, the consistent convention is $\tau = -\infty$.

Corollary 7.7 (Exact rank ratio). *For $1 \leq r \leq k$,*

$$\frac{p_r}{p_1} = \exp \left(-D \sum_{j=1}^{r-1} u_j \right).$$

This identity is local to one target distribution. It is neither a semantic correctness guarantee nor a speculative-acceptance certificate.

7.3 Distribution-free correctness is impossible

Proposition 7.8 (No universal logit-only calibrator). *Fix any measurable proposed correctness probability $q : \mathbb{R}^V \rightarrow [0, 1]$. There exist two joint laws with the same deterministic logit vector $Z = z_0$, but with correctness $Y = 1$ almost surely under one law and $Y = 0$ almost surely under the other. Consequently*

$$\max\{\text{Brier}_P(q), \text{Brier}_Q(q)\} = \max\{(1 - q(z_0))^2, q(z_0)^2\} \geq \frac{1}{4}.$$

Proof. The observation, and hence $q(Z)$, is identical in both laws, while their conditional correctness probabilities are zero and one. Either $q(z_0) \geq 1/2$ or $1 - q(z_0) \geq 1/2$, which yields the Brier bound. $\square$

This rules out distribution-free calibration without assumptions or labeled calibration. It does not prove that logits are insufficient within one fixed process, and the same adversarial construction applies to any fixed hybrid observable. Whether internal states, semantic features, or gap coordinates help is an empirical question.

8 Exact and relaxed speculative verification

Let a draft distribution q propose a token X , let $a(x) \in [0, 1]$ be its acceptance probability, and after rejection draw a replacement from h . Set

$$r = 1 - \sum_x q(x)a(x).$$

Theorem 8.1 (Acceptance/residual decomposition). *The final output follows target distribution p if and only if*

$$q(x)a(x) + rh(x) = p(x) \quad \text{for every } x. \quad (8)$$

When $r > 0$, such an h exists exactly when $q(x)a(x) \leq p(x)$ for every x , and then

$$h(x) = \frac{p(x) - q(x)a(x)}{r}.$$

When $r = 0$, exactness holds exactly when $q(x)a(x) = p(x)$ for every x , equivalently $q = p$ with full acceptance on the support of q ; acceptance at q -null points is immaterial.

Proof. Equation (8) is the law of total probability over acceptance and rejection. For $r > 0$, solving for h gives the formula. Nonnegativity is equivalent to $qa \leq p$, and the numerator sums to r , so h is normalized. If $r = 0$, the residual term vanishes, giving the stated componentwise condition. $\square$

For $q(x) > 0$, the standard maximal choice

$$a(x) = \min\{1, p(x)/q(x)\}$$

has accepted mass $\min\{p(x), q(x)\}$. It maximizes the total acceptance probability because exactness requires $qa \leq p$ and $a \leq 1$ gives $qa \leq q$. Hence

$$A_{\max} = \sum_x \min\{p(x), q(x)\} = 1 - \text{TV}(p, q). \quad (9)$$

Conditional on acceptance, when the acceptance probability is nonzero,

$$\Pr(X = x \mid \text{accepted}) = \frac{q(x)a(x)}{\sum_y q(y)a(y)}.$$

This conditional law need not itself equal p ; the residual completes the target distribution.

Now suppose a target-only gate accepts a proposed token of target rank $r \leq k$ when

$$\frac{s_1 - s_r}{D} \leq \rho.$$

Corollary 7.7 gives the transparent identity

$$\frac{p_r}{p_1} = e^{s_r - s_1} \geq e^{-\rho D}.$$

This gate omits the proposal distribution q and generally fails (8). A normalized threshold also permits an arbitrarily small ratio as D grows. Adding an absolute cap

$$s_1 - s_r \leq \varepsilon$$

ensures $p_r/p_1 \geq e^{-\varepsilon}$, but still does not make the rule exact. Lossless uses of the geometry may change draft length or verification effort while retaining the standard p/q accept-and-correct mechanism.

This distinction separates three objectives: exact sampling preserves p ; relaxed verification intentionally changes the distribution and must measure that change; semantic or judge-based decoding optimizes a different downstream criterion. Margin-aware relaxed verification and stronger acceptance certificates already exist [22, 21]; the elementary ratio identity above does not supersede them.

9 A conditional selective-risk guarantee

The coordinates do not supply calibration by themselves. One limited but exact wrapper can be stated under explicit sampling assumptions. Let $(S_i, E_i)_{i=1}^N$ be IID score/error pairs from the deployment law, fix a finite threshold grid $\mathcal{T}$ before observing them, and define

$$r_t = \Pr(E = 1 \mid S \geq t).$$

Proposition 9.1 (Simultaneous binomial risk bounds). *For every threshold meeting a predeclared minimum accepted count, compute a one-sided Clopper–Pearson upper bound at level $\alpha/|\mathcal{T}|$ [4]. With probability at least $1 - \alpha$, every reported bound simultaneously covers its corresponding r_t . On that event, any threshold selected because its bound is at most a target risk ρ satisfies $r_t \leq \rho$.*

Proof. Conditional on the number accepted at one fixed threshold, accepted errors are Bernoulli with parameter r_t . The exact one-sided bound fails with probability at most $\alpha/|\mathcal{T}|$. A union bound over the predeclared grid gives simultaneous coverage; selection among simultaneously valid bounds preserves the statement. $\square$

The assumptions are substantive. Calibration-set reuse, adaptive threshold search outside the fixed grid, dependence, and arbitrary distribution shift are not covered. More specialized conformal reasoning methods make different assumptions and targets [24]; saying only “wrap the score with conformal prediction” is not a complete method specification.

10 Independent computational and formal evidence

The universal claims above rest on proofs. Computation supplies independent implementation evidence and exact finite-instance certificates.

10.1 Independent exhaustive certificate

The checker in `proofs/independent_oracle.py` does not import the production `kaprekar` package. It uses separate integer-arithmetic implementations and exhaustively covers

$$2 \leq b \leq 8, \quad 1 \leq n \leq 7.$$

Across these 49 systems it checks 3,816,497 raw states and 973 spectra. Each case evaluates 18 obligations, including factorization, spectrum surjectivity, coefficient slack, injectivity, image equality and cardinality, multiset/fiber agreement, graph indegrees, leaf counts, conjugacy, basin partition, and hitting-time partition. Ordered raw transitions, reduced graphs, and weights are committed by SHA-256 digests.

The machine-readable artifact is `results/independent_proof_certificate.json`. Its logical certificate digest is

0aa747402f00b6901b9f11111f3dfe6066466b3425aadd455803c6373e09af75.

The certificate proves its declared finite grid, not all bases and widths.

10.2 Formalization scope

The pinned Lean project in `proofs/lean/` mechanizes the universal even/odd pairing identities, the outer-gap factorization, and the cardinality of an order-free gap multiset type. It does not yet mechanize the complete equivalence with weakly decreasing spectra, the injectivity-to-image bridge, weighted graph reconstruction, logit theorems, or speculative residual theorem. The paper therefore describes the result as partially formalized, not fully machine-checked. The independent Lean artifact accompanying the odd-base four-digit classification [3] is separate prior work.

Table 2: Evidence status of the principal claims.

Claim	Strongest evidence
Factorization, injectivity, image count	Universal proof; factorization partially formalized
Conjugacy, full graph, weighted basins, entropy	Universal proof; independent finite stress test
Decimal cycle identities and basin sizes	Exact finite certificate
Odd-base four-digit projective doubling	External theorem and external Lean artifact
Logit reconstruction, limits, and rank ratios	Universal proof/identity
Speculative target preservation condition	Universal proof
KGS improves correctness, calibration, or speed	Unproved empirical hypothesis
Production, educational, or cryptographic impact	Not established; cryptographic use contraindicated

11 Empirical protocol and falsifiable hypotheses

Although no LLM experiment is reported, the theory determines what a valid future test must control. At minimum, a preregistered benchmark should include:

- a same-capacity model on relative sorted top- k logits, plus an unrestricted sorted-logit offset control;
- u -only, D -only, (D, u) , token-identity, and tail-log-mass ablations;
- maximum probability, entropy, raw margin, conditional top- k entropy, semantic and hidden-state methods, and hybrid baselines;
- several predeclared values of k , temperature rescaling, and within-domain, cross-domain, cross-model, and prompt shifts;
- question- or sequence-disjoint train, calibration, and untouched test splits; and
- AUROC, AUPRC, Brier score, negative log-likelihood, full risk-coverage curves, AURC, paired uncertainty intervals, latency, and memory.

For approximate speculative decoding, exact and relaxed methods must appear in separate tables. Throughput must be compared at matched downstream quality or declared divergence, not at an unmatched acceptance rate.

The only empirical hypotheses retained are:

H1.

Normalized gap coordinates may provide a useful finite-sample inductive bias in regimes where local sorted-logit shape matters, but should not outperform an information-matched comparator merely by containing more information.

H2.

Adding identities, tails, semantic features, or internal probes may improve shift robustness; Proposition 7.8 does not prove that any such hybrid will succeed.

H3.

Gap diagnostics may help allocate exact verification effort or draft length. Any distribution-changing margin rule must be evaluated against MARS and exact speculative sampling.

These hypotheses are falsifiable and are not promoted to conclusions in this paper.

12 Limitations and consequences

Prior-art limitation. Outer-difference dependence is classical [20]. The contribution is the explicit injective code and its graph-theoretic consequences, not the original discovery of the coordinate idea.

Finite versus universal evidence. The exhaustive certificate covers exactly its stated grid. Decimal basin shares are finite fixed-width results and change if the input law or leading-zero policy changes. The universal claims depend on the displayed proofs; only a subset is currently formalized in Lean.

Representation loss. Sorting drops token identity, top- k truncation drops tail detail, and normalization drops scale unless D is retained. Even (D, u) drops the common offset, which can matter outside shift-invariant observation models. The representation becomes ill-conditioned near flat spectra and changes with k .

No empirical LLM result. There are no real-model logits, correctness labels, distribution shifts, latency measurements, or human evaluations in this work. No state-of-the-art, production-safety, speedup, or educational-impact claim is supported.

Operational consequences. For finite-dynamics research, the reduction replaces exponential raw enumeration with an exact polynomial fixed-base graph and preserves raw basin weights. For security, the same collapse is a warning against using the map as a hash or entropy mechanism. For LLM decision systems, false acceptance can expose users to incorrect answers while false rejection adds cost and withholds useful answers; both must be measured at a declared risk-coverage point. For speculative decoding, a target-only margin can be useful as a diagnostic but cannot silently be relabeled exact sampling.

13 Conclusion

The durable mathematical object behind 6174 is an ordered-gap law. Its explicit linear reconstruction is superincreasing, which yields an exact one-step image count and a conjugacy between the post-first-step dynamics and a polynomial-size reduced graph at fixed base. Fiber weights then recover the complete functional graph, raw basins, depth histograms, and entropy evolution. Exact decimal computations demonstrate why cycles, not only fixed points, are essential to the real dynamics.

Normalized adjacent gaps also provide a precise coordinate system for sorted top- k logits. Their value is representational and diagnostic: they separate spread from relative shape and expose exact rank ratios. Their limitations are equally exact: they add no information relative to relative sorted logits, are singular at flat spectra, omit tail detail, and cannot universally calibrate correctness. Likewise, target-only margin acceptance is not a substitute for the accept-and-residual condition required to preserve a target distribution. This combination of positive structure and negative limits is the paper’s intended practical contribution.

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Code and data availability

Source code, tests, Lean files, machine-readable result artifacts, and the arXiv source package are available at <https://github.com/thewisecrab/kaprekar-geometry>. The repository contains no human-subject or proprietary dataset. The original compiled manuscript is preserved separately from this major revision, and exact artifact and source-tree digests are recorded in `results/manifest.json`. The absence of an empirical LLM dataset is intentional and is a limitation, not missing evidence for a performance claim.

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